

Facial Attribute Prediction: Eigenfaces vs. CNN vs. ResNet50

Contributor - Collaborated with two of my friends

Overview

We explored three approaches to predicting demographic attributes—age, race, and gender—from face images, using Kaggle’s UTKface dataset. The goal was to compare a classical computer vision (Eigenfaces/PCA) against two deep convolutionary deep learning approaches (CNN and ResNet50), and see how each one compares in accuracy, training cost, and generalization.

Dataset

- UTKFaces: ~20k+ labeled face images, each tagged with age, gender, and race
- Images resized to 64x64 for all models
 - Grayscale versions for PCA
 - Maintained RGB for CNN and ResNet50
- Explored class balance and age distribution before modeling. We used class weighting for race prediction to offset any imbalance across the 5 race categories.

Methods

1. **PCA** Flattened grayscale images and reduced dimensionality with PCA, then used KNN and Logistic/Linear Regression on top of the reduced features for each task.
2. **Convolutional Neural Network** Built a three-layer CNN from scratch (Conv2D → MaxPooling, X3) for each task. We then used a binary classification for gender, multi-class for race, and regression for age.
3. **ResNet50** Used ImageNet-pretrained ResNet50 as a feature extractor with a custom classification head to see whether pretrained features would outperform a small CNN trained from scratch.

Each model was evaluated with confusion matrices, precision-recall (ROC) curves, and MAE/MSE plus a predicted-vs-actual scatter(age regression).

Results

Task	PCA + KNN	PCA + Logistic/Linear	Custom CNN	Resnet50
Gender (accuracy)	74.2%	84.4%	89.2%	72.2%
Race (accuracy)	53.5%	65.8%	75.8%	30.9%
Age (MAE)	12.35	11.41	6.84	-

The custom CNN outperformed both the classical PCA and the ResNet50 across all three tasks. ResNet50 underperformed because it was frozen and pretrained on 224x224 ImageNet images, so its features didn't transfer cleanly to our 64x64 inputs. We notice that this may have not been the best idea and unfreezing later layers and using larger input sizes may increase performance.

Tools Used

Python, OpenCV, scikit-learn (PCA, KNN, Logistic/Linear Regression, evaluation metrics), TensorFlow/Keras (CNN, ResNet50), Pandas, Matplotlib, Seaborn

Learnings

- We found that the race classification was the hardest task across all methods for underrepresented classes. This signifies that dataset class imbalance materially affect real-world model fairness and performance.